



BACKEND DEVELOPER

155 HAND WRITTEN
INTERVIEW QUESTIONS
AND ANSWERS

API

API

Backend Basics (1-20)

1. What is backend development?

Ans: Backend development handles server logic, databases, and APIs.

2. What is a server?

Ans: A system that responds to client requests.

3. What is client-server architecture?

Ans: Client sends requests, server responds.

4. What is REST API?

Ans: An API that follows REST principles using HTTP.

5. What is HTTP?

Ans: Protocol for data communication on the web.

6. Common HTTP methods?

Ans: GET, POST, PUT, DELETE, PATCH.

7. What is statelessness?

Ans: Each request is independent.

8. What is JSON?

Ans: Lightweight data exchange format.

9. What is middleware?

Ans: Code that runs between request and response.

10. What is MVC architecture?

Ans: Model, View, Controller pattern.

11. What is monolithic architecture?

Ans: Entire app as one unit.

12. What is microservices architecture?

Ans: App split into small independent services.

13. What is backend framework?

Ans: Tool to simplify backend development.

14. Examples of backend languages?

Ans: Java, Python, Node.js, PHP, Go.

15. What is Node.js?

Ans: JavaScript runtime for server-side.

16. What is Express.js?

Ans: Node.js web framework.

17. What is Django?

Ans: Python backend framework.

18. What is Spring Boot?

Ans: Java backend framework.

19. What is API versioning?

Ans: Managing API changes.

20. What is environment variable?

Ans: Stores config data securely.

Database Concepts (21-45)

21. What is a database?

Ans: Stores structured data.

22. Types of databases?

Ans: SQL and NoSQL.

23. What is SQL?

Ans: Structured Query Language.

24. Examples of SQL databases?

Ans: MySQL, PostgreSQL, Oracle.

25. What is NoSQL?

Ans: Non-relational database.

26. Examples of NoSQL databases?

Ans: MongoDB, Redis, Cassandra.

27. Difference between SQL and NoSQL?

Ans: Structured vs flexible schema.

28. What is primary key?

Ans: Unique record identifier.

29. What is foreign key?

Ans: Links two tables.

30. What is indexing?

Ans: Improves query performance.

31. What is normalization?

Ans: Reduces data redundancy.

32. What is denormalization?

Ans: Improves read performance.

33. What is ACID?

Ans: Atomicity, Consistency, Isolation, Durability.

34. What is transaction?

Ans: Group of database operations.

35. What is JOIN?

Ans: Combines tables.

36. Types of JOINS?

Ans: INNER, LEFT, RIGHT, FULL.

37. What is ORM?

Ans: Maps objects to database tables.

38. Examples of ORM?

Ans: Hibernate, Sequelize, SQLAlchemy.

39. What is migration?

Ans: Database schema changes.

40. What is schema?

Ans: Database structure.

41. What is query optimization?

Ans: Improving query speed.

42. What is sharding?

Ans: Splitting data across servers.

43. What is replication?

Ans: Copying data to multiple servers.

44. What is Redis?

Ans: In-memory data store.

45. What is caching?

Ans: Storing data for faster access.

API & Authentication (46-75)

46. What is API?

Ans: Interface for communication between systems.

47. What is RESTful API?

Ans: API following REST rules.

48. What is SOAP?

Ans: XML-based API protocol.

49. Difference between REST and SOAP?

Ans: REST is lightweight, SOAP is strict.

50. What is GraphQL?

Ans: Query-based API language.

51. What is API Gateway?

Ans: Manages API requests.

52. What is authentication?

Ans: Verifying user identity.

53. What is authorization?

Ans: Checking user permissions.

54. Difference between auth and authz?

Ans: Identity vs access control.

55. What is JWT?

Ans: JSON Web Token.

56. What is OAuth?

Ans: Authorization protocol.

57. What is session-based auth?

Ans: Uses server-side sessions.

58. What is token-based auth?

Ans: Uses tokens like JWT.

59. What is refresh token?

Ans: Renews access token.

60. What is hashing?

Ans: One-way encryption.

61. What is salting?

Ans: Adding randomness to hashes.

62. What is bcrypt?

Ans: Password hashing algorithm.

63. What is HTTPS?

Ans: Secure HTTP.

64. What is SSL/TLS?

Ans: Encryption protocols.

65. What is CORS?

Ans: Cross-origin access control.

66. What is rate limiting?

Ans: Restricting API requests.

67. What is API throttling?

Ans: Controls traffic flow.

68. What is idempotency?

Ans: Same request gives same result.

69. What is API documentation?

Ans: Explains API usage.

70. What is Swagger?

Ans: API documentation tool.

71. What is Postman?

Ans: API testing tool.

72. What is request validation?

Ans: Checking input data.

73. What is response status code?

Ans: Server response indicator.

74. Common HTTP status codes?

Ans: 200, 201, 400, 401, 404, 500.

75. What is payload?

Ans: Data sent in request.

Security & Performance (76-100)

76. What is SQL injection?

Ans: Database attack using queries.

77. What is XSS?

Ans: Cross-site scripting attack.

78. What is CSRF?

Ans: Cross-site request forgery.

79. How to prevent SQL injection?

Ans: Use prepared statements.

80. How to prevent XSS?

Ans: Input sanitization.

81. What is firewall?

Ans: Network security system.

82. What is encryption?

Ans: Protecting data.

83. What is symmetric encryption?

Ans: Same key for encrypt and decrypt.

84. What is asymmetric encryption?

Ans: Public and private keys.

85. What is load balancing?

Ans: Distributes traffic.

86. What is scalability?

Ans: Handling increased load.

87. Horizontal vs vertical scaling?

Ans: Add servers vs upgrade server.

88. What is latency?

Ans: Response delay.

89. What is throughput?

Ans: Requests per second.

90. What is performance tuning?

Ans: Optimizing system speed.

91. What is memory leak?

Ans: Unreleased memory.

92. What is garbage collection?

Ans: Memory cleanup process.

93. What is logging?

Ans: Recording application events.

94. What is monitoring?

Ans: Tracking system health.

95. What is health check API?

Ans: Checks service status.

96. What is CDN?

Ans: Content delivery network.

97. What is compression?

Ans: Reducing response size.

98. What is gzip?

Ans: Compression method.

99. What is timeout?

Ans: Request time limit.

100. What is failover?

Ans: Backup system activation.

System Design & DevOps (101-130)

101. What is system design?

Ans: Designing scalable backend systems.

102. What is high availability?

Ans: Minimal downtime.

103. What is consistency?

Ans: Same data everywhere.

104. What is CAP theorem?

Ans: Consistency, Availability, Partition tolerance.

105. What is message queue?

Ans: Async communication system.

106. Examples of message queues?

Ans: RabbitMQ, Kafka.

107. What is Kafka?

Ans: Distributed event streaming platform.

108. What is asynchronous processing?

Ans: Non-blocking execution.

109. What is cron job?

Ans: Scheduled task.

110. What is background job?

Ans: Runs outside request cycle.

111. What is Docker?

Ans: Containerization tool.

112. What is container?

Ans: Isolated app environment.

113. What is Kubernetes?

Ans: Container orchestration tool.

114. What is CI/CD?

Ans: Automated build and deployment.

115. What is Git?

Ans: Version control system.

116. What is branching?

Ans: Parallel development.

117. What is merge conflict?

Ans: Code collision issue.

118. What is environment setup?

Ans: Configuring dev, test, prod.

119. What is reverse proxy?

Ans: Routes client requests.

120. What is Nginx?

Ans: Web server and proxy.

121. What is Apache server?

Ans: Web server.

122. What is cloud computing?

Ans: On-demand computing resources.

123. Examples of cloud providers?

Ans: AWS, Azure, GCP.

124. What is serverless?

Ans: No server management.

125. What is AWS Lambda?

Ans: Serverless function service.

126. What is auto-scaling?

Ans: Automatic resource adjustment.

127. What is blue-green deployment?

Ans: Zero downtime deployment.

128. What is canary release?

Ans: Gradual rollout.

129. What is rollback?

Ans: Reverting deployment.

130. What is infrastructure as code?

Ans: Managing infra using code.

Testing & Best Practices (131-155)

131. What is backend testing?

Ans: Testing server-side logic.

132. Types of testing?

Ans: Unit, integration, system.

133. What is unit testing?

Ans: Testing small code units.

134. What is integration testing?

Ans: Testing combined components.

135. What is API testing?

Ans: Testing endpoints.

136. What is mock testing?

Ans: Simulating dependencies.

137. What is test coverage?

Ans: Code tested percentage.

138. What is clean architecture?

Ans: Separation of concerns.

139. What is DRY principle?

Ans: Do not repeat yourself.

140. What is SOLID principle?

Ans: OOP design principles.

141. What is dependency injection?

Ans: Injecting dependencies.

142. What is code review?

Ans: Reviewing code quality.

143. What is documentation?

Ans: Explaining code usage.

144. What is error handling?

Ans: Managing failures.

145. What is exception?

Ans: Runtime error.

146. What is custom error?

Ans: User-defined error.

147. What is validation layer?

Ans: Checks input rules.

148. What is pagination?

Ans: Dividing data into pages.

149. What is sorting?

Ans: Ordering data.

150. What is filtering?

Ans: Selecting specific data.

151. What is backend optimization?

Ans: Improving speed and efficiency.

152. What is production-ready code?

Ans: Stable and secure code.

153. What makes a good backend developer?

Ans: Logic, scalability, security focus.

154. What is API security best practice?

Ans: Auth, validation, rate limits.

155. What is real-world backend challenge?

Ans: Handling scale, failures, and security.